HDOC DAY-69

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BATCH-15

Aim

To write a C program to generate the first n Fibonacci numbers using recursion and count the total number of recursive function calls made.

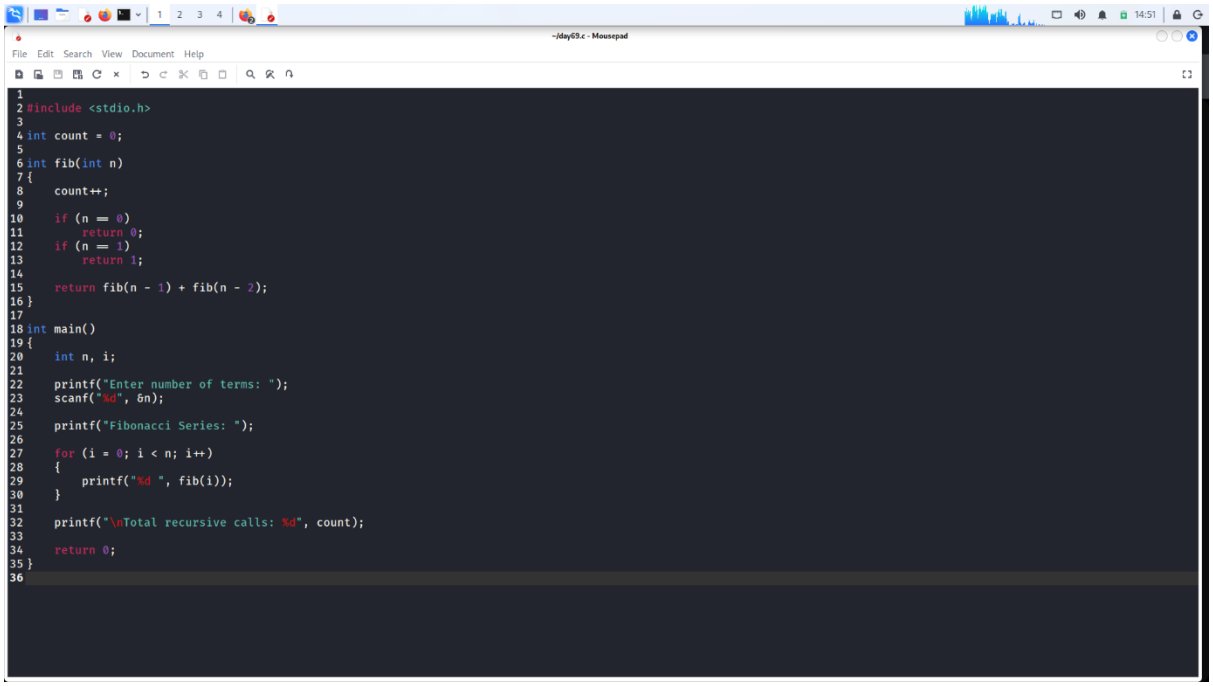
Problem Statement

Write a recursive function to generate the first n Fibonacci numbers. Also count the number of recursive calls made.

Algorithm

1. Start the program.
2. Declare a global variable count = 0.
3. Define a recursive function fib(n).
4. Increment count whenever the function is called.
5. If n == 0, return 0.
6. If n == 1, return 1.
7. Otherwise, return fib(n-1) + fib(n-2).
8. Read the number of terms n from the user.
9. Use a loop to print the first n Fibonacci numbers.
10. Display the total number of recursive calls.
11. Stop the program.

>C PROGRAM:



```c
#include <stdio.h>

int count = 0;

int fib(int n)
{
    count++;

    if (n == 0)
        return 0;
    if (n == 1)
        return 1;

    return fib(n - 1) + fib(n - 2);
}

int main()
{
    int n, i;

    printf("Enter number of terms: ");
    scanf("%d", &n);

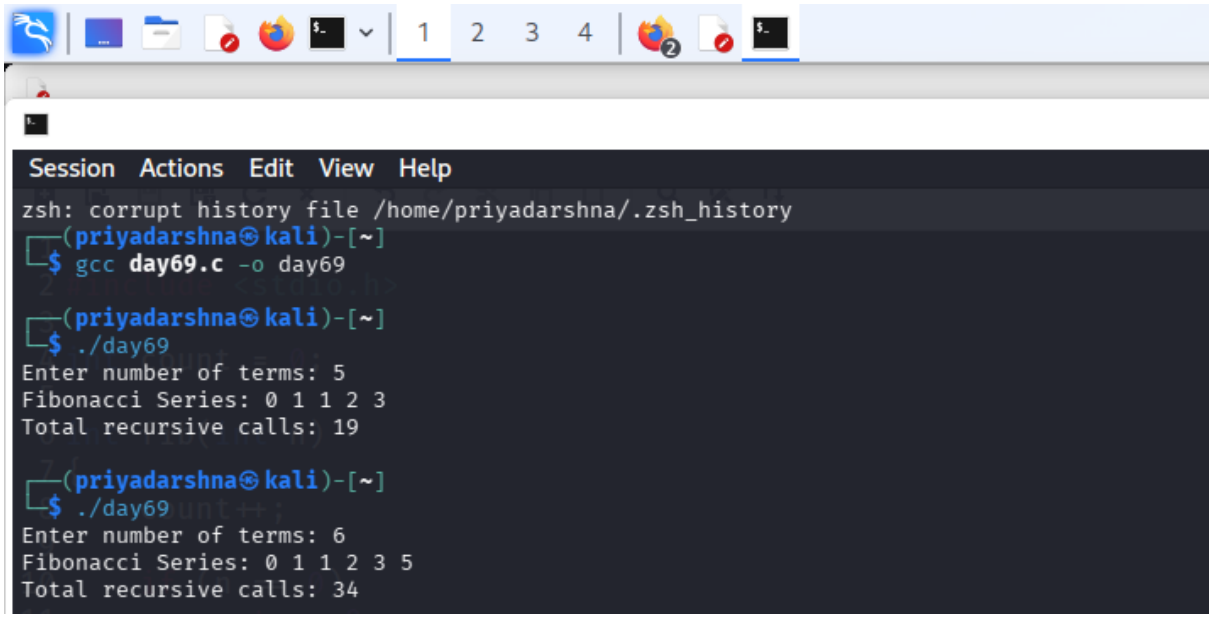
    printf("Fibonacci Series: ");

    for (i = 0; i < n; i++)
    {
        printf("%d ", fib(i));
    }

    printf("\nTotal recursive calls: %d", count);

    return 0;
}
```

>OUTPUT:



```
Session  Actions  Edit  View  Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
┌──(priyadarshna㉿kali)-[~]
└─$ gcc day69.c -o day69

┌──(priyadarshna㉿kali)-[~]
└─$ ./day69
Enter number of terms: 5
Fibonacci Series: 0 1 1 2 3
Total recursive calls: 19

┌──(priyadarshna㉿kali)-[~]
└─$ ./day69
Enter number of terms: 6
Fibonacci Series: 0 1 1 2 3 5
Total recursive calls: 34
```